

Lecture 13

Discrete Probability Distribution

Some Discrete Probability Distribution:

1. Binomial distribution
2. Geometric distribution
3. Poisson distribution
4. Hyper-geometric distribution
5. Negative binomial distribution
6. Discrete Uniform distribution

Some Continuous Probability Distribution:

1. Normal distribution
2. Exponential distribution
3. Gamma distribution
4. Beta distribution
5. Lognormal distribution
6. Weibull distribution

1 Binomial Distribution

If we repeat an experiment **n** times with probability of success at a single trial **P**, then the probability of exactly **x** success out of **n** trials is

$$P(x) = {}^n C_x \cdot p^x \cdot (1 - p)^{n-x}; \quad x = 0, 1, 2, \dots, n$$

Where,

n = total number of trials

x = number of success

p = probability of success in a single trial

1 - p = probability of failure in a single trial

Conditions for Binomial Distribution

1. Each trial has only two outcomes (often called 'Success' and 'Failure')
2. The trials are independent
3. The probability of outcomes does not change
4. The number of trials is fixed
5. The variable of interest is the number of successful trials (x) among (n) trials

Application of Binomial Distribution

It has already been mentioned that the experimental results of random experiment which can be classified into two classes: success (S) and failure (F), are expressed by binomial probability law. Therefore, the results of the following experiments can be expressed by binomial probability distribution:

1. to identify an industrial product as defective or non-defective,
2. identification of present or absent employees in a factory,
3. identification of skilled or unskilled labourers in an industry,
4. to hit the target successfully,
5. to study the successful implementation of a project,
6. to conduct the opinion survey supporting or opposing an issue,
7. to study the effectiveness of a medicine or a chemical,
8. to identify a children as boy or girl,

Example 1:

Suppose you are rolling a die repetitively. Let's say getting 6 is called success. Find the probability that you'd be successful exactly twice after rolling the die 7 times?

Solution:

n = total number of trials = 7

x = number of success = 2

p = probability of getting 6 = 1/6

$$\begin{aligned}P(x) &= n_{C_x} \times p^x \times (1-p)^{n-x} \\&= {}^7C_2 \times \left(\frac{1}{6}\right)^2 \times \left(1 - \frac{1}{6}\right)^{7-2} \\&= {}^7C_2 \times \left(\frac{1}{6}\right)^2 \times \left(1 - \frac{1}{6}\right)^5 \\&= 0.234\end{aligned}$$

Example 2:

You are appearing at 6 consecutive exams. Probability for you to pass each of the exam, $P = \frac{4}{5}$. What is the probability that you pass exactly 4 of total 6 exams?

Solution:

n = total number of trials = 6

x = number of success = 4

p = probability of pass = 4/5

$$\begin{aligned}P(x) &= n_{C_x} \times p^x \times (1-p)^{n-x} \\&= {}^6C_4 \times \left(\frac{4}{5}\right)^4 \times \left(1 - \frac{4}{5}\right)^{6-4} \\&= {}^6C_4 \times \left(\frac{4}{5}\right)^4 \times \left(1 - \frac{4}{5}\right)^2 \\&= 0.246\end{aligned}$$

Example 3:

Suppose, A and B are two wrestlers. In a single match both of them have an equal chance of winning and there is no tie in this form of wrestling. They are fighting each other in a 6-match tournament, Y = the number of matches won by A. find the distribution of Y.

Solution:

p = probability of A winning a match

(1-p) = probability of B winning a match

Since chances are both equal for A & B,

$$p = 1-p$$

$$2p = 1$$

$$p = 1/2$$

From binomial distribution we know that, probability of A to win exactly 'y' matches in a 'n' match tournament,

$$P(y) = {}^n C_y \cdot p^y \cdot (1-p)^{n-y}$$

Here n = 6,

y = 0, 1, 2, 3, 4, 5, 6

$$P(Y = 0) = {}^6 C_0 \times \left(\frac{1}{2}\right)^0 \times \left(1 - \frac{1}{2}\right)^{6-0} = \frac{1}{2^6}$$

$$P(Y = 1) = {}^6 C_1 \times \left(\frac{1}{2}\right)^1 \times \left(1 - \frac{1}{2}\right)^{6-1} = \frac{3}{2^5}$$

$$P(Y = 2) = {}^6 C_2 \times \left(\frac{1}{2}\right)^2 \times \left(1 - \frac{1}{2}\right)^{6-2} = \frac{15}{2^6}$$

$$P(Y = 3) = {}^6 C_3 \times \left(\frac{1}{2}\right)^3 \times \left(1 - \frac{1}{2}\right)^{6-3} = \frac{5}{2^4}$$

$$P(Y = 4) = {}^6 C_4 \times \left(\frac{1}{2}\right)^4 \times \left(1 - \frac{1}{2}\right)^{6-4} = \frac{15}{2^6}$$

$$P(Y = 5) = {}^6 C_5 \times \left(\frac{1}{2}\right)^5 \times \left(1 - \frac{1}{2}\right)^{6-5} = \frac{3}{2^5}$$

$$P(Y = 6) = {}^6 C_6 \times \left(\frac{1}{2}\right)^6 \times \left(1 - \frac{1}{2}\right)^{6-6} = \frac{1}{2^6}$$

So, the distribution of Y:

Y	0	1	2	3	4	5	6
P(Y)	$1/2^6$	$3/2^5$	$15/2^6$	$5/2^4$	$15/2^6$	$3/2^5$	$1/2^6$

Mean of Binomial Random Variable:

$$Mean = E(x) = np$$

Variance of Binomial Random Variable:

$$\sigma^2 = Var(x) = np(1-p)$$

Standard Deviation of Binomial Random Variable:

$$\sigma(x) = \sqrt{Var(x)} = \sqrt{np(1-p)}$$

Example 4:

Let the probability of a student taking STA 201 = 0.25, 30 students are randomly chosen from the students of BRACU.

- Find the probability that exactly 8 students out of 30 took STA201.
- Probability that fewer than 5 students out of 30 took STA201.
- Calculate the mean and standard deviation of this binomial distribution.

Solution:

a) Here, $n = 30$, $p = 0.25$, $x = 8$

$$P(X = 8) = {}^{30}C_8 \times (0.25)^8 \times (1 - 0.25)^{30-8} = 0.1593$$

b) $P(X < 5) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) = 0.0979$

c) $Mean = np = 30 \times 0.25 = 7.5$

$$Var(x) = np(1 - p) = 30 \times 0.25 \times 0.75 = 5.625$$

$$Standard\ deviation = \sqrt{5.625} = 2.372$$

Example 5:

If 95% of households have a TV and 8 houses are surveyed, what is the probability that more than 6 have a TV? [Ans: 0.943]

Example 6:

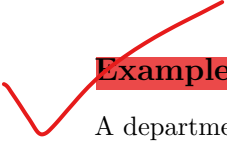
There are 3 multiple choice questions in a MCQ test. Each MCQ consists of four possible choices and only one of them is correct. If an examinee answers those MCQ randomly (without knowing the correct answers)

- What is the probability that exactly any two of the answers will be correct? [0.141]
- What is the probability that at least two of the answers will be correct? [0.157]
- What is the probability that at most two of the answers will be correct? [0.985]
- What will be the average or expected number of correct answers? [0.75]
- Also, find the standard deviation of number of correct answers. [0.75]

Example 7:

Six biased coins are tossed once. From previous experience it is known that H:T = 2:3. Find the probability that

- There will be 4 heads. [0.1382]
- There will be at best 3 heads. [0.8208]
- There will be at least 2 heads. [0.7667]
- There will be no head. [0.0467]
- There will be 3 tails. [0.2765]
- If the coins are unbiased, what would be the probabilities of the above events? [0.0586, 0.6563, 0.8906, 0.0156, 0.3125]



Example 8:

A departmental store takes the responsibility of supply of their commodities in the customer's house. Usually, in 95% cases the store does its daily duty perfectly. Find the probability that, in a randomly selected week

1. Three days the store fails to do the job perfectly. [0.003564]
2. At best 2 days the store fails to do the job perfectly. [0.9962]
3. If the job performance is observed for 52 weeks, then in how many weeks the store fails 2 days per week to do the job perfectly? [2 weeks]
4. At least three days the store do the job perfectly. [0.9999]

2 Geometric Distribution

If we repeat an experiment until success occurs where probability to succeed in a single experiment is p , then the require number of trials to be x has probability

$$P(x) = (1 - p)^{x-1} \cdot p; \quad x = 1, 2, 3 \dots, \infty$$

Where,

x = total number of trials

$x-1$ = number of unsuccessful trials

p = probability of success in a single trial

$1 - p$ = probability of failure in a single trial

Conditions for Geometric Distribution

1. Each trial has only two outcomes
2. The trials are independent
3. The probability of outcomes does not change
4. The variable of interest is the number of trials until the first success

Mean of Geometric Random Variable:

$$Mean = E(x) = \frac{1}{p}$$

Variance of Geometric Random Variable:

$$\sigma^2 = Var(x) = \frac{1-p}{p^2}$$

Standard Deviation of Geometric Random Variable:

$$\sigma(x) = \sqrt{Var(x)} = \frac{\sqrt{1-p}}{p}$$

Example 9:

Suppose you are rolling a die repetitively. Let's say rolling a die 6 is called success and you stop rolling the die right after the first success. Find the probability that it'd take you exactly 7 turns to be successful.

Solution:

Here, P = probability of getting 6 = $1/6$

x = total number of trials = 7

$$\begin{aligned} P(x) &= (1 - p)^{x-1} \times p \\ &= \left(1 - \frac{1}{6}\right)^{7-1} \times \left(\frac{1}{6}\right) \\ &= \left(1 - \frac{1}{6}\right)^6 \times \left(\frac{1}{6}\right) \end{aligned}$$

Example 10:

You are appearing at 6 consecutive exams. Probability for you to pass each of the exam, $p = 4/5$. Once you pass an exam, there is no need to sit for the rest. What is the probability that for passing the exam you have to sit for all 6 exams?

Solution:

Here, x = number of trials = 6

p = probability of passing in exam = $4/5$

$$\begin{aligned} P(x) &= (1 - p)^{x-1} \times p \\ &= \left(1 - \frac{4}{5}\right)^{6-1} \times \left(\frac{4}{5}\right) \\ &= \left(1 - \frac{4}{5}\right)^5 \times \left(\frac{4}{5}\right) \end{aligned}$$

Example 11:

2% of all tires produced by company ABC has a defect. A random sample of hundred tires is tested for quality assurance.

- a) What is the probability that the 6th tire selected is the first to have a defect?
- b) What is the probability that the first defect is identified among the first 3 samples?
- c) How many tires would you expect to test until you find the first defective one?
- d) Calculate the variance and standard deviation?

Solution:

a) probability that a tire is defective, $p = 2\% = 0.02$

$$P(x = 6) = (1 - 0.02)^5 \times 0.02 = 0.0181$$

b)

$$\begin{aligned} P(X \leq 3) &= P(X = 1) + P(X = 2) + P(X = 3) \\ &= (0.98^0 \times 0.02) + (0.98^1 \times 0.02) + (0.98^2 \times 0.02) \\ &= 0.0588 \end{aligned}$$

c) $E(x) = \frac{1}{0.02} = 50$

d) $\text{var}(x) = \frac{1-p}{p^2} = \frac{0.98}{0.02^2} = 2450$

3 Poisson Distribution

Let, X = a random variable usually counts or number of occurrences

Then, X is a Poisson random variable with distribution function (pmf)

$$P(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}; \quad x = 0, 1, 2, \dots$$

Example: Some random quantities that can be modeled by Poisson distribution:

- Number of car accidents daily in a city.
- Number of patients in a waiting room in an hour.
- Number of surgeries performed in a month.
- Number of website visitors per month.
- Number of defective items produced by a machine per hour.

Mean of Poisson Random Variable:

$$\text{Mean} = E(x) = \lambda$$

Variance of Poisson Random Variable:

$$\sigma^2 = \text{Var}(x) = \lambda$$

Standard Deviation of Poisson Random Variable:

$$\sigma(x) = \sqrt{\text{Var}(x)} = \sqrt{\lambda}$$

Note:

- λ is the average (mean) of the poisson distribution.
- If X = The number of patients seen in the emergency unit in a day, and if $X \sim \text{Poisson}(\lambda)$, then:
- The average (mean) of patients seen every day in the emergency unit = λ
- The average (mean) of patients seen every month in the emergency unit = 30λ
- The average (mean) of patients seen every year in the emergency unit = 365λ
- The average (mean) of patients seen every hour in the emergency unit = $\frac{\lambda}{24}$

Example 12:

A small business receives, on average, 12 customers per day. What is the probability that the business will receive exactly 8 customers in a day?

Solution: $\lambda = 12$

$$P(X = x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$
$$P(X = 8) = \frac{e^{-12} \cdot 12^8}{8!} = 0.066$$

Example 13:

A student receives on average 7 text message in a two hours period.

- a) What is the probability that the student will receive exactly 9 text messages in two hours?
- b) What's the probability that the student will receive exactly 24 text messages in 8 hours?

Solution:

- a) Here, $\lambda = 7$, $x = 9$

$$P(X = 9) = \frac{e^{-7} \cdot 7^9}{9!} = 0.101$$

- b) Here, $\lambda = 28$, $x = 24$

$$P(X = 24) = \frac{e^{-28} \cdot 28^{24}}{24!} = 0.06$$

Example 14:

A small business receives on average 8 calls per hour

- a) probability that it will receive 7 calls in an hour = ?
- b) what is the probability that the business will receive, at most 5 calls in an hour?
- c) What is the probability that the business will receive more than 6 calls in an hour?

Solution:

- a) Here, $\lambda = 8$, $x = 7$

$$P(X = 7) = \frac{e^{-8} \cdot 8^7}{7!} = 0.1396$$

- b)

$$\begin{aligned} P(X \leq 5) &= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) \\ &= e^{-8} \times \left[\frac{8^0}{0!} + \frac{8^1}{1!} + \frac{8^2}{2!} + \frac{8^3}{3!} + \frac{8^4}{4!} + \frac{8^5}{5!} \right] \\ &= 0.191 \end{aligned}$$

- c)

$$\begin{aligned} P(X > 6) &= 1 - P(X \leq 6) \\ &= 1 - e^{-8} \times \left[\frac{8^0}{0!} + \frac{8^1}{1!} + \dots + \frac{8^6}{6!} \right] \\ &= 0.687 \end{aligned}$$

Example 15:

The average number of errors on a page of a certain magazine is 0.2. What is the probability that the next page (or a randomly selected page) you read contains

- i. 0 (zero) error?
- ii. 2 or more errors?
- iii. What is the average error per page?
- iv. Also, find standard deviation of the number of errors.

Example 16:

Suppose that the number of accidents per day in a city has a Poisson distribution with average 2 accidents.

1. What is the probability that in a day:
 - (i) The number of accidents will be 5?
 - (ii) The number of accidents will be less than 2?
2. What is the probability that there will be 6 accidents in 2 days
3. What is the probability that there will be no accidents in an hour